22 . BIRD-FLY OR NOT

AIM

To write a Prolog program to determine whether a particular bird can fly or not using facts, rules, and queries.

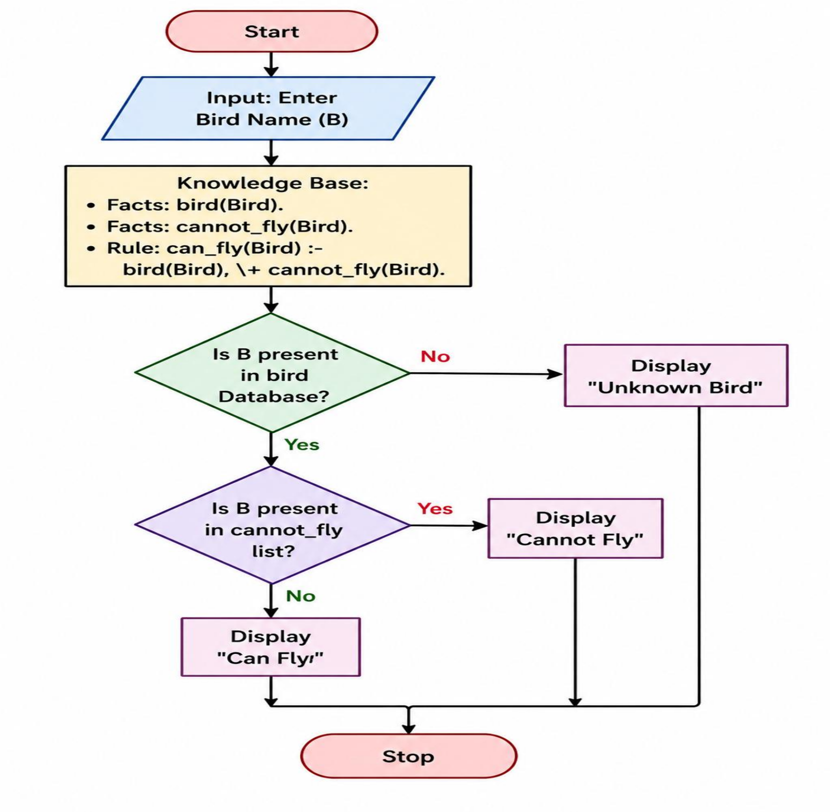
OBJECTIVE

- To store bird information as Prolog facts.
- To identify birds that cannot fly.
- To use a rule to determine the flying ability of a bird.
- To execute queries for checking whether a particular bird can fly.

ALGORITHM

1. Start.
2. Define the list of birds as facts.
3. Define birds that cannot fly.
4. Create the rule can_fly(Bird).
5. Enter the name of a bird as a query.
6. Check whether the bird is present in the database.
7. If the bird is not in the cannot_fly list, return true.
8. Otherwise, return false.
9. Display the result.
10. Stop.

FLOWCHART



CODE:

```
bird(eagle).

bird(parrot).

bird(pigeon).

bird(sparrow).

bird(penguin).

bird(ostrich).

cannot_fly(penguin).

cannot_fly(ostrich).

can_fly(Bird) :-
    bird(Bird),
    \+ cannot_fly(Bird).
```

OUTPUT

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File  Edit  Settings  Run  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/New/Documents/Prolog/bird fly.pl compiled 0.00 sec, 9 clauses
?-  can_fly(eagle).
true.

?-  can_fly(penguin).
false.

?- 
```

RESULT:

Thus, the Prolog program was successfully implemented to determine whether a particular bird can fly or not, and the required queries produced the expected results.